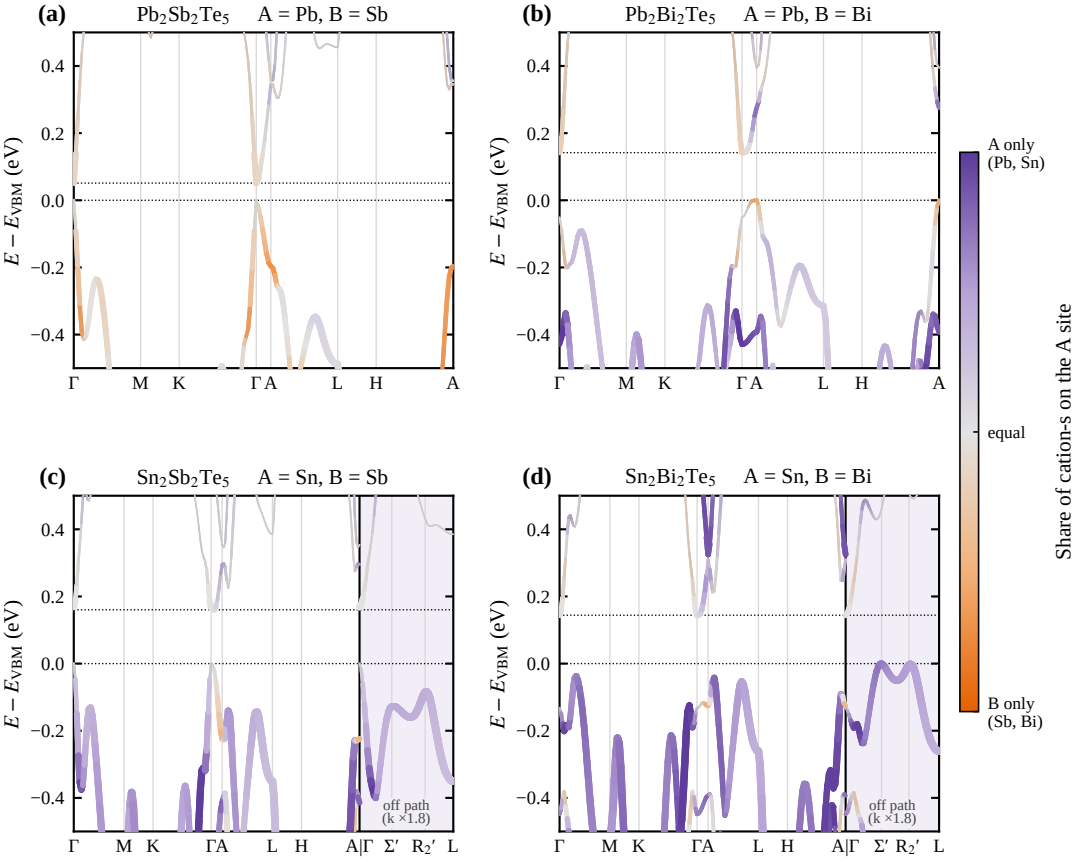

(a)
$Pb_2Sb_2Te_5$ A = Pb, B = Sb
(b)
$Pb_2Bi_2Te_5$ A = Pb, B = Bi
(c)
$Sn_2Sb_2Te_5$ A = Sn, B = Sb
(d)
$Sn_2Bi_2Te_5$ A = Sn, B = Bi
$E - E_{VBM}$ (eV)
0.4
0.2
0.0
−0.2
−0.4
Γ M K Γ A L H A
Γ M K ΓA L H A|Γ Σ′ R2′ L
off path (k ×1.8)
A only (Pb, Sn)
equal
B only (Sb, Bi)
Share of cation-s on the A site